

# Latent Space Augmentation

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## Introduction

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## Literature Review

Recent advances in protein language models (PLMs) have transformed how protein sequences are represented and analyzed. Inspired by natural language processing, models such as ESM and other transformer-based architectures are trained on millions of protein sequences and learn contextual embeddings that capture structural and functional properties without requiring explicit biological annotations. A comprehensive review of protein language models demonstrated that learned protein embeddings consistently improve downstream prediction tasks, including function prediction, localization, and structural analysis (Chen et al., 2025). These findings suggest that PLMs capture biologically meaningful sequence patterns, providing a powerful foundation for representation learning in protein-related machine learning applications.

Building on these representations, autoencoders and variational autoencoders (VAEs) have been utilized to learn a lower-dimensional latent space that preserves critical protein characteristics. Hawkins-Hooker et al. (2020) demonstrated that VAEs can generate novel protein variants while maintaining functional properties, while Riesselman et al. (2018) showed that latent and generative models can capture mutational constraints and predict the effects of sequence variations. Additional work indicated that proteins with similar functions often cluster within nearby regions of latent space, suggesting that these learned representations capture meaningful biological structure rather than simple sequence similarity (Sevgen et al., 2023).

However, data augmentation remains challenging in protein classification tasks because direct amino acid mutations can alter biologically important motifs and disrupt the relationship between protein sequence and its functional labels. In other domains, latent space augmentation has been shown to generate diverse training examples while preserving higher-level semantic information. Recent protein studies have similarly demonstrated that latent representations learned from PLMs can support generation and optimization tasks, suggesting that biologically meaningful transformations are possible within learned latent spaces (Shin et al, 2019).

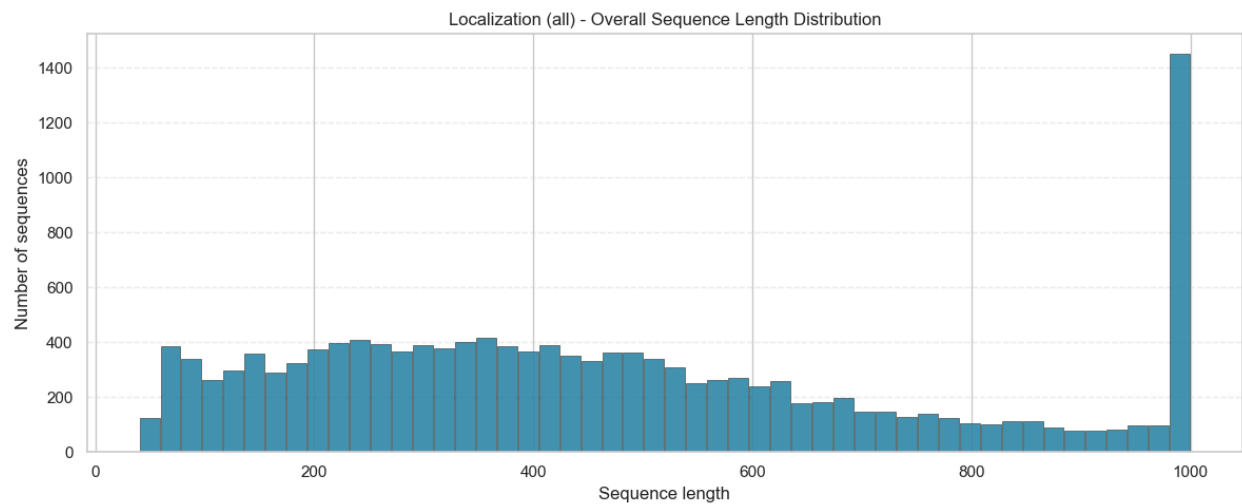
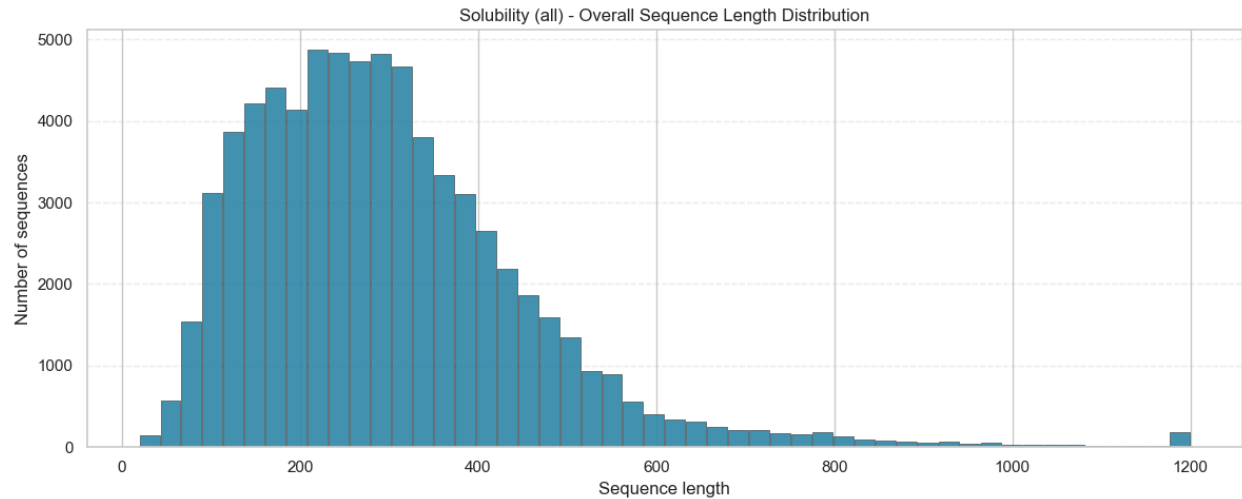
Despite these advances, relatively little work has examined whether perturbations applied within learned protein latent spaces can serve as a safer augmentation strategy than direct sequence mutations for downstream classification tasks. This study addresses this gap by comparing mutation-based augmentation and autoencoder-based latent-space perturbation across protein classification datasets. Ultimately, the goal is to determine whether latent space augmentation better preserves biologically relevant information while improving downstream predictive performance.

### Exploratory Data Analysis

Prior to the augmentation experiments, we conducted exploratory data analysis (EDA) on the processed PEER solubility and subcellular localization datasets to assess dataset quality, label structure, sequence-length distributions, amino acid composition, and potential risks for downstream augmentation. [PLACEHOLDER -> Overall, EDA suggested that solubility would be an easier task to train on].

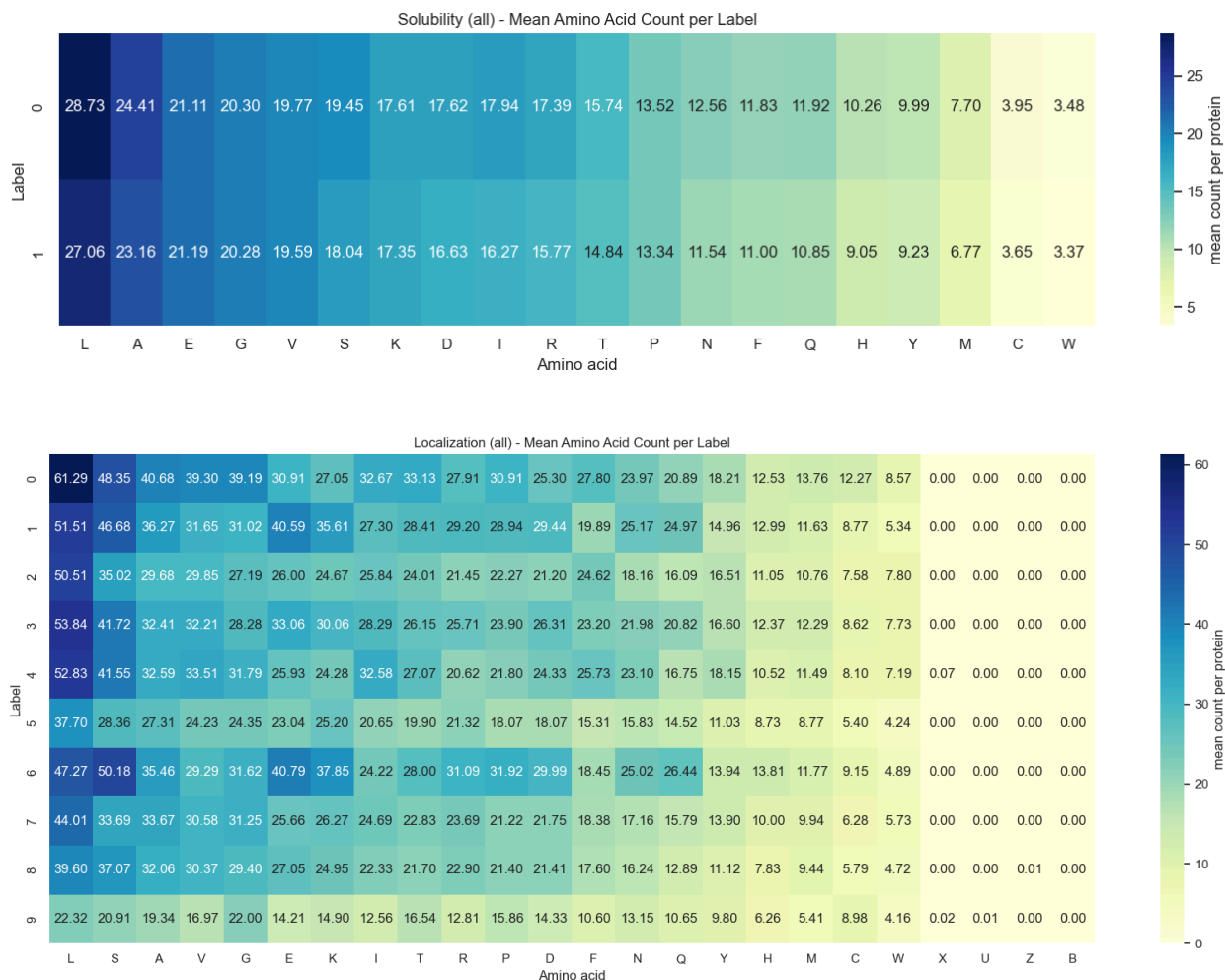
The solubility task is a binary classification problem with significantly more training data than localization. The processed solubility split contains 62,478 training sequences, 6,942 validation sequences, and 1,999 test sequences, with a median sequence length of 275 amino acids and a maximum length of 1,200 amino acids. The training split is moderately balanced, with approximately 58% of examples in class 0 and 42% in class 1, and the validation split follows nearly identical proportions. The test set is close to perfectly balanced.

*Sequence length differed meaningfully between the two tasks.* Solubility sequences were shorter on average, with an all-split mean length of 298.4 amino acids and median length of 275 amino acids. Localization sequences were longer, with an all-split mean length of about 472 amino acids and median length of 421 amino acids. The solubility length distribution was right-skewed, with many proteins near 200-300 amino acids and a long tail extending beyond 1,000 amino acids. Localization appeared more bimodal, with one concentration near 300-400 amino acids and another near 950-1,000 amino acids. When training, we found that sequences that were too long led to complications during training. [PLACEHOLDER - haven't fully implemented here yet -> These patterns are relevant for batching, padding, truncation, autoencoder reconstruction difficulty, and the computational cost of embedding or decoding full sequences.] Especially for training the autoencoder, a more uniform distribution of training size was a key driving in the decision to train on the solubility dataset.

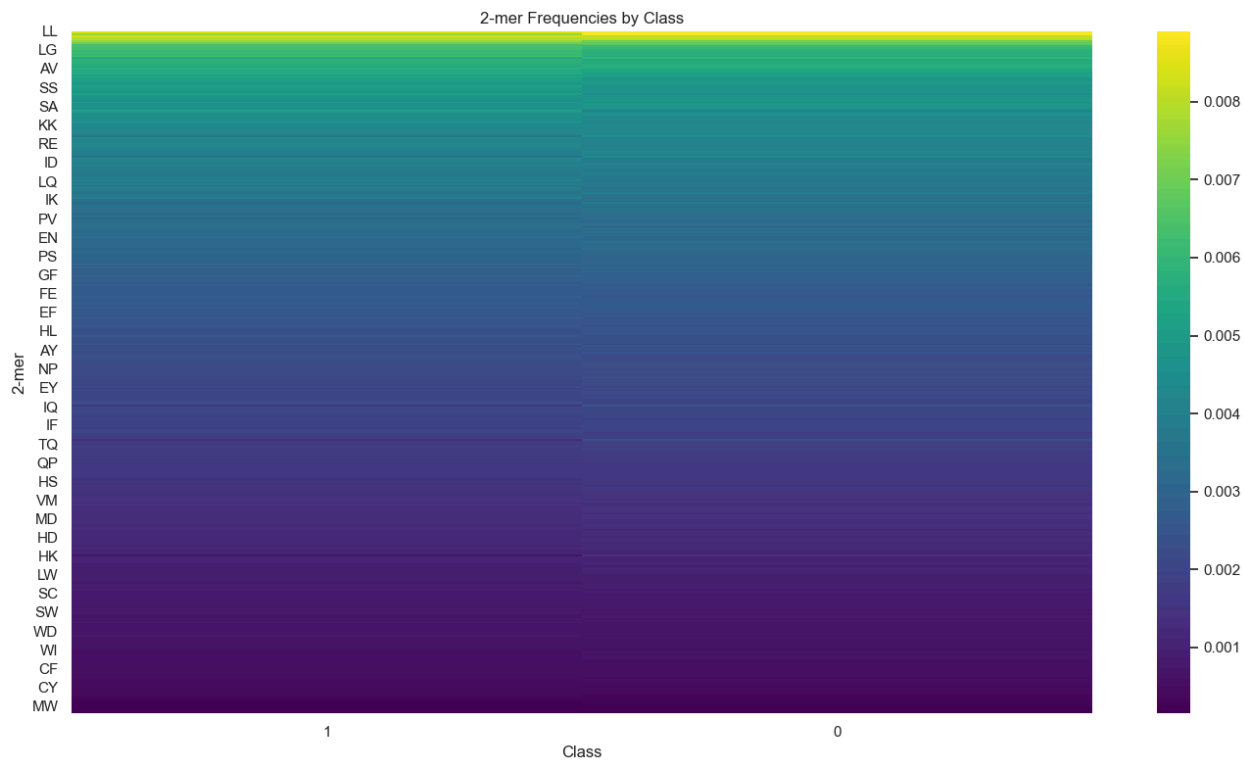


Amino acid composition analysis showed that the most common residues were broadly similar across tasks, but the degree of class-level variation differed. In solubility, the most frequent training residues were L, A, E, G, V, S, K, and I, and mean residue counts by class appeared similar. This suggests that single-residue composition alone may not strongly separate soluble and insoluble proteins, even though biochemical properties such as charged, polar, and hydrophobic residue fractions may still contribute to solubility. In localization, the most frequent residues were L, S, A, E, K, G, V, and P, and class-level composition differences were more visible, suggesting that residue

composition may contain stronger label signals for localization. Ambiguous or uncommon amino acid symbols were not present within the solubility training data but appeared rarely in localization, including X, U, and B.



The k-mer analysis indicated that short local sequence patterns may be more informative than individual amino acid counts. In particular, 2-mers appeared more interpretable and less sparse than 3-mers or 4-mers, especially for the multiclass localization task. This indicates that when evaluating reconstruction from latent space, looking at 2-mers rather than individual amino acid composition may be a better indication of preservation of biological significance.



Methods

Results

Discussion

Conclusion

## References

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